

Fraud Transaction Report

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Introduction.

Situation Complication Quetsion

Variables and their values.

This will give an indication of the level of variation in each variable. This also may provide insights that will guide the focus of the analysis.

Customer ID has 1092 unique values. It may generally not be useful to analyse this variable except for very specific purpose.

Items	Unique values
customer_id	1,092
merchant_category	12
billing_country	10
shipping_country	10
email_domain	8
channel	4
currency	3
product_type	3
payment_method	3
is_cross_border	2

Missing values

The device_fingerprint has the highest number of missing values, followed by the merchant risk scores.

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Variables	Missing values
device_fingerprint	185
merchant_risk_score	87
email_domain	41
txn_count_1h	19
txn_count_24h	12
transaction_id	0
transaction_date	0
customer_id	0
customer_age	0
customer_tenure_days	0
is_vip	0
merchant_id	0
merchant_category	0
amount	0
currency	0

Variables	Missing values
payment_method	0
channel	0
card_present	0
device_type	0
billing_country	0
shipping_country	0
ip_country	0
is_cross_border	0
is_new_device	0
prev_chargebacks_012m	
product_type	0
actual_fraud	0
predicted_fraud	0

Thirteen (13) Business Questions

In this section, I explored some business-related questions, provided evidence-based answers and explained their implications for business performance and decision-making.

Fraud rate and losses.

(1) Total fraudulent transaction

What proportion of total transactions are fraudulent? The overall fraud rate for the month is about 2% of the total transaction. Though this varies vary by currency, with EUR recording the highest fraud rate at 1.5%, compared with GBP and USD.

EUR accounted for the majority of fraudulent transactions, with an average fraudulent transaction value of approximately EUR 60. The total value of fraudulent transactions was approximately 1,200 EUR, 429 GBP, and 73 USD for EUR, GBP, and USD transactions, respectively.

Actual fraud	Count	Proportion
Legitimate	1,352	98%
Fraudulent	25	2%

(2) Fraud incidence by currency

Are there differences in total fraud rate and losses reported by currency?

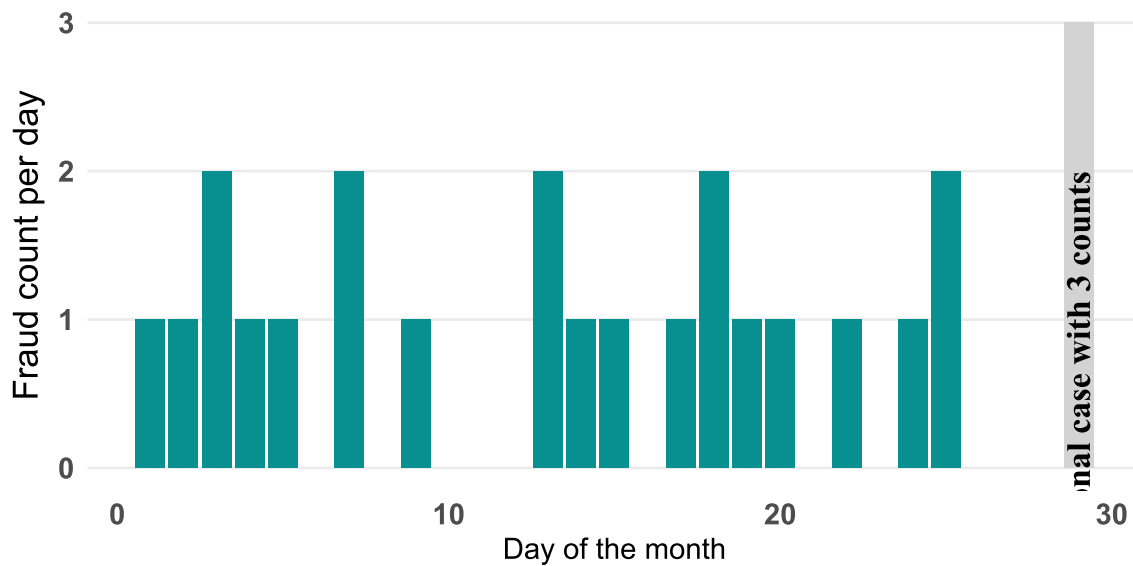
Fraud incidence by currency				
Currency	Fraud count	Total fraud loss	Average fraud loss	Fraud rate
EUR	20	1,183.28	59.2	1.5
GBP	4	428.74	107.2	0.3
USD	1	72.84	72.8	0.1

(3) Daily fraud rate

What is the daily fraud rate?

One or two cases of fraud incidence per day are typical, with three cases on a single day being an outlier. Fraud incidence in a given month ranges between one and two per day with three being an outlier.

On any day, when fraud occurs, **one** or **two** cases are typical. In a given month, fraud may not occur every day.



Customer risk analysis

(4) Customer age and risk

Which customer age groups have the highest incidence of fraud? Customer risk analysis: Fraud incidence does not appear to vary substantially by customer age. Customers across all age groups appear to be similarly likely to experience transaction fraud.

The three largest fraudulent transactions by transaction value involved card payments. However, further analysis is required to determine whether card payments are more susceptible to fraud than other payment methods.

Customer age	Fraud count
79	1
78	1
77	1
73	1
68	1
65	1
64	1
62	1
59	1
56	1
55	1
54	1
49	1
42	2
41	1
36	1
34	1
28	1
26	2
25	1
24	1
21	1
19	1

(5) Vip status and risk

Are VIP customers more or less likely to experience fraud than non-VIP customers?

No transactions involving VIP customers were identified as fraudulent; all fraudulent transactions involved non-VIP customers.

Vip status	currency	Fraud count	Total fraud loss
0	EUR	20	1,183.28
0	GBP	4	428.74
0	USD	1	72.84

(6) Customer tenure and risk

Is customer tenure associated with a lower risk of fraud? Fraud incidence appears to vary by customer tenure. Customers with tenure below the average are more likely to have experienced fraud.

Specifically, 60% of customers with shorter tenure were victims of fraud, compared with 40% of customers with longer tenure.

Tenure category	Fraud count	Proportion
Short tenure	15	60%
Long tenure	10	40%

(7) Customers exposure profile?

Which five customers experienced the highest fraud losses? It is also evident that the highest-value fraud involved card payments. Does this suggest that card payments are more susceptible to fraud?

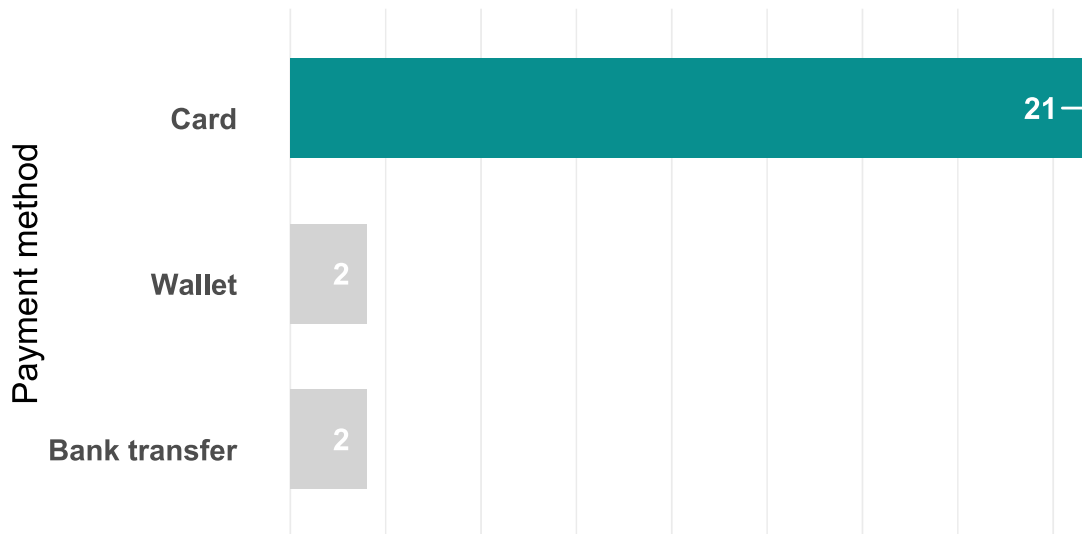
Customer ID	Customer age	Amount	Currency	Channel	Payment	Tenure
C000804	24	216.80	EUR	mobile_app	card	3,378
C000581	59	193.91	GBP	web	card	607
C001866	77	118.64	GBP	web	card	1,620
C001576	42	118.49	EUR	pos	card	671
C001779	54	103.16	EUR	web	card	772

Digital properties and fraud incidence: ### (8) Payment methods and risk? Which payment methods have the highest incidence of fraud?

Card payment transactions recorded the highest fraud.

Card payment transactions recorded the highest fraud cases.

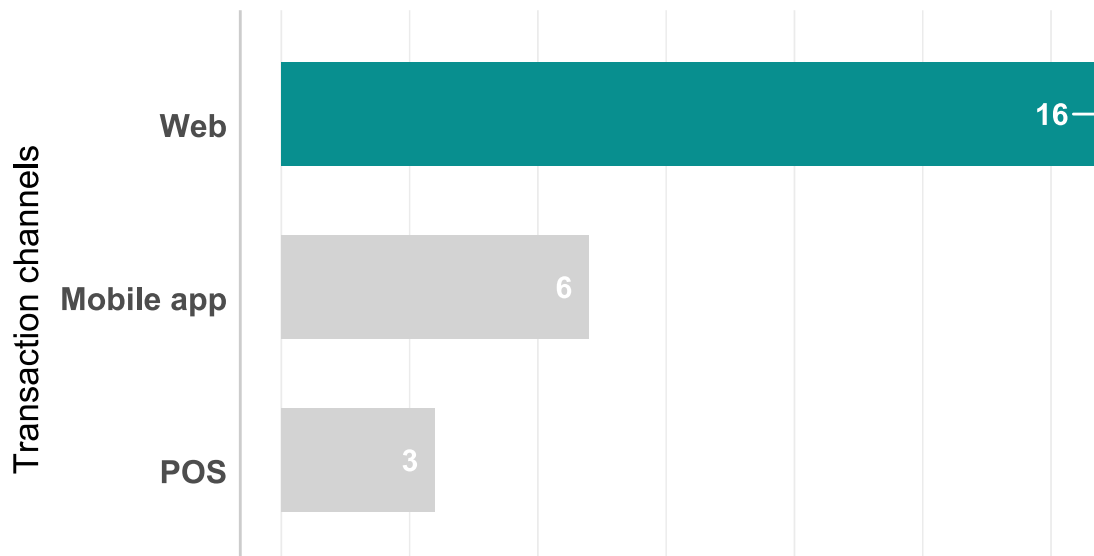
Making it the riskiest transaction payment method.



(9) Transactions channels risk?

Which transactions channels have the highest fraud rates? Web channel transactions reported the highest fraud.

Web transactions channel recorded the highest fraud cases, making it the most vulnerable transaction channel.



Geographic locations and fraud patterns

(10) Cross-border transactions and fraud patterns

Are cross-border transactions riskier? First what are cross-border transactions? I defined them as those where the customer's ip_country, billing_country, and shipping_country differ.

The majority of fraudulent transactions (88%) are cross-border transactions. These are transactions in which the IP country differs from the billing country and the billing country differs from the shipping country.

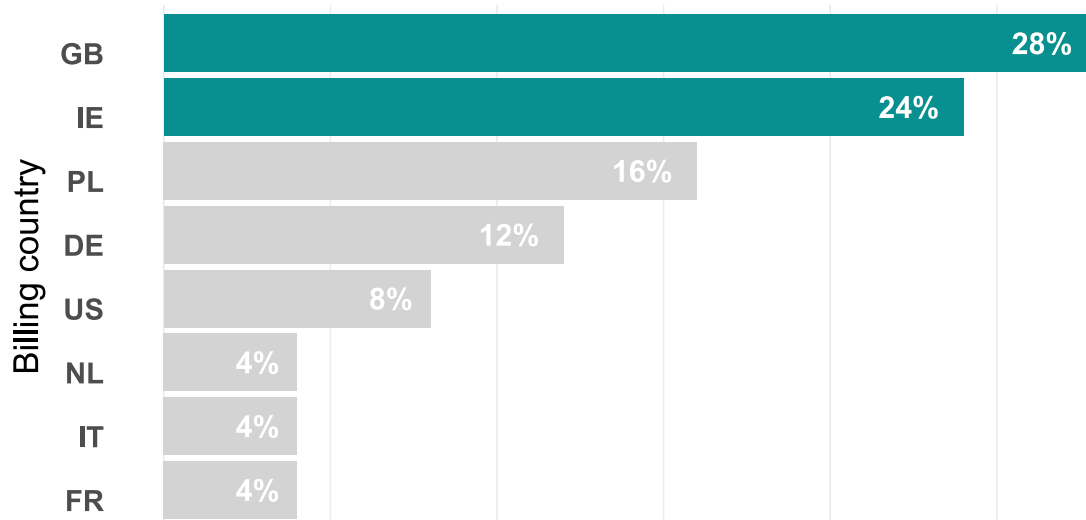
Status	Nature	Fraud count	Prop.
Fraudulent	cross border	22	88%
Fraudulent	within border	3	12%

(11) Billing destination and risk

Which are the top billing countries for fraudulent transaction?

Great Britain (GB) and **Ireland (IE)** are the top billing destinations for fraudulent transactions.

They represent more than 50% of the total fraud losses.



Merchant Risk Analysis

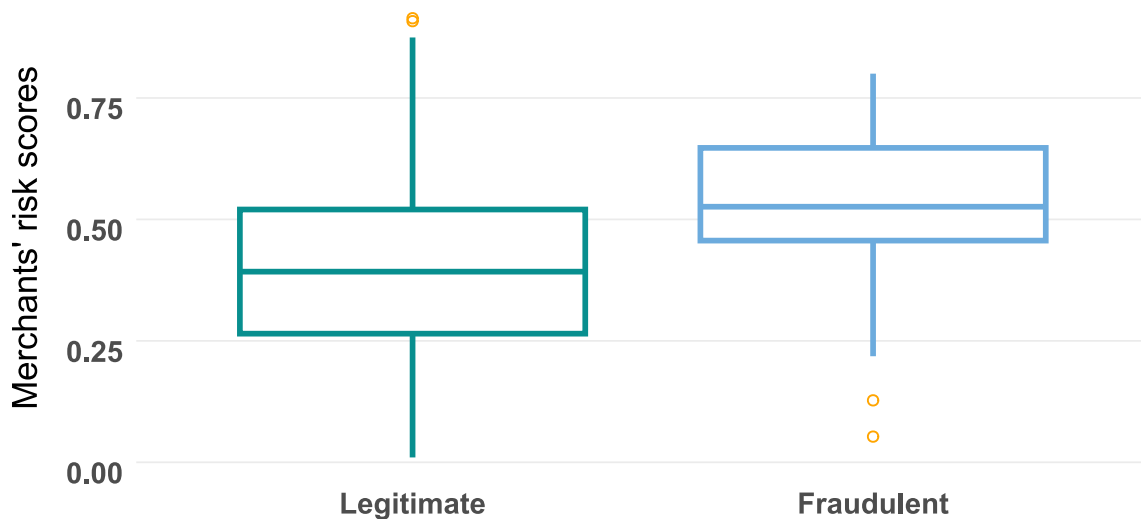
(12) Fraud incidence and merchant risk

Does fraud incidence vary by merchant risk score?

Fraudulent transactions are common with high-risk merchants relative to non-fraudulent ones than their low-risk counterparts.

Fraudulent transactions are more prevalent with high-risk merchants.

Low-risk merchants are less vulnerable.



(13) Merchant categories and risk.

Which merchant categories have the highest fraud rates?

Marketplaces and **fashion** merchants are the most susceptible to fraudulent transactions.

